

Ferry Rachman Kurniawan

Jakarta, Indonesia · 0821-2523-0439 · Kurniawan.ferry.rach@gmail.com · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFILE

Computer Engineering graduate from Bina Nusantara University, specializing in IoT and embedded systems. Skilled in designing end-to-end IoT solutions — from ESP32/Arduino hardware prototyping to cloud backends with MQTT and REST APIs. Experienced in integrating embedded devices, mobile applications, and cloud services into scalable systems, with a focus on real-time communication, multi-layer security design, and cross-platform mobile development using Flutter.

EDUCATION

Bina Nusantara University 2022 – 2026
Bachelor of Computer Engineering · GPA 3.38 · *Finalist Gemastik 2025* · *Top 50 APAC Google Solution Challenge 2025* · *Special Winner Binus IoT Expo 2024* · *Best Microcontroller Final Project*

EXPERIENCE

Zegen Solusi Mandiri | Mobile Programmer – Internship Feb 2025 – Jul 2025

- Engineered 2+ cross-platform mobile applications for Android and iOS using Flutter, contributing to a production-grade codebase.
- Delivered 10+ features and bug fixes across multiple sprints, measurably improving app stability and reducing crash rates.
- Collaborated with cross-functional teams to maintain 100% on-time feature delivery across all sprint cycles.

PROJECTS

Smart Medicine Dispenser | Full Stack Developer [\[Mobile\]](#) [\[ESP\]](#)
Tech: *Flutter, Golang, Azure, ESP32, Arduino, FCM, MQTT, REST API*

- Architected an end-to-end IoT medicine dispenser integrating mobile app, Azure cloud, and ESP32 hardware into a unified system.
- Engineered real-time device-cloud sync via MQTT protocol, enabling sub-second status updates and remote control.
- Designed 5+ mobile features — scheduling, monitoring, FCM push notifications, and reminder — for seamless user experience.

LockGuard Pro – Smart IoT Lock | IoT & Mobile Developer
Tech: *ESP32, GNSS u-blox NEO-M8, OV2640 Camera, Flutter, Firebase, Cloud AI*

- Engineered a smart lock system integrating 5+ hardware components (ESP32, GNSS, camera, solenoid, TFT LCD) into a single cohesive device.
- Implemented 3-layer authentication (PIN, QR code, face recognition), eliminating single-point-of-failure access vulnerabilities.
- Delivered geofencing-based unlocking with GNSS real-time validation and a Flutter app providing live monitoring and access history.

LEADERSHIP & ORGANIZATION

Bina Nusantara English Club (BNEC)

- Manager of IT (2024): Revamped the official website and digital platforms; conducted web development training for 20+ internal staff members.
- Vice Project Operational, New Member Recruitment (2023): Directed 2 divisions to execute a large-scale campaign, digitizing registration workflows.

TECHNICAL SKILLS

Languages: Dart · PHP · C/C++ **Frameworks:** Flutter · Laravel · Qt · Arduino **Hardware:** ESP32 · STM32 · ATmega328P
Tools & Cloud: Git · MQTT · REST API · Firebase · Azure · Golang · Fusion 360 **Soft Skills:** Leadership · Analytical Thinking · Adaptability